

Explainable BiLSTM-MHA-Based IDS for IoT Using SHAP and Zero-Day Attack Detection

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Abstract—Because of their distributed nature, resource limitations, and the growing sophistication of cyberthreats, Internet of Things (IoT) environments pose serious security issues. The heterogeneity of devices and the vast volume of data they generate make IoT networks particularly vulnerable to a wide range of attacks, posing serious risks to privacy, safety, and critical infrastructure. This paper presents a new intrusion detection system (IDS) architecture utilizing bidirectional long short-term memory with multi-head attention (BiLSTM-MHA) to improve the detection of cyberattacks in IoT networks. The model was trained and evaluated using the comprehensive TON_IoT dataset, encompassing multiple attack classes and normal traffic. Our approach achieved a high accuracy of 98.79%, a low loss of 0.0349, and a false positive rate (FPR) of just 0.0391%, while maintaining an efficient inference time of 0.002 seconds per instance, rendering it appropriate for real-time implementation in IoT systems. To mitigate the issue of class imbalance, the Focal Loss technique was employed, yielding superior performance compared to the baseline model trained on the imbalanced dataset. Furthermore, to enhance model transparency and interpretability, Explainable AI (XAI) methods were integrated using SHapley Additive exPlanations (SHAP), enabling a detailed interpretation of the model's decision-making process and the identification of the most influential features associated with each attack class. This provided deeper insights into the model's behavior and validated the importance of key features in detecting different intrusion types. We further evaluated the model's generalization capability through a Zero-Day attack assessment using a Leave-One-Attack-Out (LOAO) strategy. The model was trained without one attack class and tested on that unseen class to simulate a realistic zero-day scenario. It achieved a testing accuracy of 97.91%, a loss of 0.0591, and an F1-score of 98.94%, confirming its robustness and adaptability to unknown threats.

Index Terms—IDS, BiLSTM, MHA, Cyber-threats, Focal loss, XAI, SHAP.

I. INTRODUCTION

The evolution of the IoT has redefined how modern systems interact with the physical world, enabling ubiquitous connectivity across smart homes, healthcare systems, industrial automation, and critical infrastructure. This vast interconnected ecosystem generates massive volumes of data and supports real-time communication between heterogeneous devices, often with minimal human intervention [1]. However, the same features that make IoT systems powerful; decentralization, scalability, and openness; also make them

inherently vulnerable to cyber threats. Unlike traditional IT systems, IoT devices frequently operate with constrained computational resources and limited security provisions [2], making them easy targets for adversaries. Furthermore, the diversity in communication protocols and hardware platforms introduces inconsistencies in security enforcement, creating blind spots in the network that attackers can exploit [3]. The consequence is a growing surface area for attacks, ranging from data breaches and device manipulation to large-scale Distributed Denial-of-Service (DDoS) campaigns [4]. The problem is that some attacks alter their behavior (Zero-Day attacks), rendering them invisible. Protecting these IoT networks requires intelligent, adaptive, and explainable security solutions capable of detecting not only known attack patterns but also emerging threats [5]. IDS have become an essential element of IoT security frameworks [6], aiming to identify malicious behavior by analyzing traffic patterns and device activities [7]. However, conventional IDS solutions, including signature-based and shallow machine learning (ML) models, often fall short in this domain. Signature-based approaches struggle to detect novel or evolving threats [8], while ML methods suffer from poor generalization to unseen attacks and offer limited transparency into their decision-making processes. The need for robust, adaptive, and explainable IDS frameworks in IoT environments has never been more urgent; particularly in the face of zero-day attacks that cannot be captured by traditional rule sets [9]. Addressing these challenges requires models that can understand temporal and contextual dependencies in IoT traffic data, adapt to novel threats, and provide interpretability to support operational trust and debugging. Deep learning (DL) architectures, especially recurrent neural networks (RNN), offer a promising direction, but their black-box nature and sensitivity to unseen patterns still pose limitations when deployed in real-world scenarios [9]. Motivated by these concerns, this work presents an intelligent and explainable intrusion detection framework tailored for IoT networks. Our approach integrates a BiLSTM network with a MHA mechanism to effectively learn both sequential dependencies and complex feature interactions in network traffic data. To enhance the transparency of the detection process, we employ SHapley Additive exPlanations (SHAP) [10],

enabling granular interpretation of model predictions and highlighting the most influential features behind each detection outcome. This not only supports security analysts in understanding attack behavior but also increases confidence in the system's decisions. In response to the growing need for secure, interpretable, and zero-day resilient IDS solutions in IoT ecosystems, this research makes the following key contributions:

- Design an innovative IDS architecture that integrates BiLSTM and MHA to proficiently capture both temporal dependencies and contextual interactions in IoT network traffic data.
- Demonstrate the model's ability to generalize and detect previously unseen (zero-day) attacks named Leave-One-Attack-Out (LOAO) strategy, addressing a major limitation in traditional and signature-based IDS approaches.
- Employ SHAP to provide interpretable insights into model decisions, allowing identification of key features contributing to attack detection and enhancing trust in the system.
- Conduct extensive experiments using the TON_IoT dataset, which includes a wide range of realistic attack scenarios, to validate the performance, generalizability, and interpretability of the proposed IDS model.
- Compare the proposed BiLSTM-MHA model against conventional ML and DL approaches, showing superior performance in terms of accuracy, precision, recall, F1-score, and robustness to novel threats.

This paper's remaining sections are arranged as follows: The problem statement is presented in Section II, which also examines relevant research from current literature. The suggested technique and the preprocessing procedures used on the dataset are described in depth in Section III. Section IV presents the experimental setup, discusses the obtained results, and provides a critical analysis. Section V wraps up the work and suggests possible lines of inquiry for further study.

II. RESEARCH GAP AND LITERATURE REVIEW

The IoT explosive expansion has brought a vast array of networked gadgets into daily life, from smart homes and healthcare systems to industrial control networks. While these systems offer improved automation and efficiency, their heterogeneous nature, resource limitations, and often weak security configurations make them highly susceptible to cyber-attacks. IDSs have emerged as a critical line of defense for identifying malicious behavior in IoT networks. However, conventional IDS models often suffer from several key limitations:

- **Low generalizability;** Many models do well against known attacks, but they are unable to identify emerging or novel risks (zero-day attacks).
- **High FPR;** leading to unnecessary alerts and reduced trust in the system.
- **Poor interpretability;** the majority of IDSs based on DL function as "black boxes," providing little insight into the decision-making process.

- **Inadequate real-time performance;** existing solutions may be too slow or resource-intensive for deployment in latency-sensitive or low-power IoT environments.

In this context, numerous state-of-the-art studies have sought to address these challenges through the development of advanced methodologies and detection frameworks, among these, we have the study by A. Alqahtani [11] proposes an optimized deep autoencoder combined with BiLSTM model for intrusion detection in IoT-Fog computing environments. The approach enhances feature extraction and sequence learning to detect cyber threats effectively. The study added two new metrics named Cohen's Kappa coefficients and Mathew correlation. Extensive experiments on CICIDS2017 nad BoT-IoT benchmark datasets demonstrate improved accuracy and reduced time. In [12], S. Bi et al. present an intrusion detection model for the Power IoT (PIoT) that combines a Deep Belief Network (DBN) with a BiLSTM network. Experimental results show that the hybrid model performs with an accuracy of 98.24% and a computation time of 4.53 seconds. Another work by R. Nagarajan [13] proposes a BiLSTM model for network intrusion detection in Software-Defined Networking-IoT (SDN-IoT) environments. Experimental evaluations on CSE-CIC-IDS2018 and SDN-IoT benchmark datasets achieved an accuracy of 99.96% and 92.90% , respectively.

III. PROPOSED METHODOLOGY

This section presents the data preprocessing methodology and the architecture of the proposed framework, which leverages the combined strengths of the BiLSTM-MHA model.

A. Data Preparation

- **Label Encoding for Classification:** The type column (which contains attack categories) is mapped to numerical values from 0 to 9, converting multiclass text labels into integers.
- **Feature Selection:** Selects relevant input features (predictorNames) to be used as predictors (input variables). These features include numeric values, IPs, protocol data, DNS/SSL/HTTP attributes, etc.
- **Encoding Categorical Features:** Categorical (string-based) features are converted to numeric values using `LabelEncoder` for compatibility with machine learning models.
- **Data splitting:** The selected TON_IoT dataset, after pre-processing, is divided into two distinct subsets.

- 1) **Training:** The suggested model was trained using 80% of the pre-processed data, guaranteeing reliable model creation and assessment.
- 2) **Validation:** 20% of the pre-processed data was reserved to validate the performance of the BiLSTM-MHA model, allocated for training our proposed model, ensuring robust model development and evaluation.

- **One-Hot Encoding for Labels:** Since this is a multiclass classification problem with 10 classes, the labels are one-hot encoded (Prepare for softmax classification)

B. Model architecture:

The proposed model is a hybrid DL framework that successfully captures contextual and temporal correlations in the input data by fusing the advantages of MHA and BiLSTM methods. The structure of the architecture is as follows:

- **Input Layer:** The model accepts input sequences of shape (43, 1), where 43 represents the number of features per instance, treated as time steps to leverage sequential modeling.

- **Bidirectional LSTM Layer:** The input sequence is processed both forward and backward using a bidirectional LSTM layer, which has 64 units in each direction. This enables the model to concurrently record past and future dependencies, resulting in an output of shape (43, 128).

- **Multi-Head Attention Layer:** The output of the BiLSTM is passed through a MHA layer with 16 attention heads and a key dimensionality of 64. This mechanism allows the model to attend to different positions in the sequence and capture important contextual relationships.

- **Layer Normalization:** The output of the attention layer is subjected to a Layer Normalization operation in order to stabilize and speed up the training process.

- **Global Average Pooling:** The normalized output is then passed through a Global Average Pooling layer, which reduces the temporal dimension by computing the average across all time steps, yielding a fixed-length vector representation.

- **Dropout Layer:** A dropout layer with a rate of 0.5 is applied for regularization, mitigating overfitting by randomly disabling a portion of the neurons during training.

- **Output Layer:** Finally, a probability distribution across 10 output classes; corresponding to various network attack types; is generated using a fully linked Dense layer with a softmax activation function.

Categorical cross-entropy loss is used to train the model, which is suitable for multi-class classification problems, once it has been assembled using the Adam optimizer.

IV. EXPERIMENTS, OUTCOMES, AND ANALYSIS

A. TON_IoT Dataset exploration

TON_IoT¹: Telemetry, Operating systems, and Network data for the Internet of Things) dataset, developed by the Cyber Range Lab of UNSW Canberra, which is specifically designed for research in IoT and cybersecurity. The TON_IoT dataset integrates telemetry data from operating system logs, network traffic, and IoT sensors, making it a comprehensive and realistic benchmark for intrusion detection and threat classification tasks. It contains a rich set of connection features extracted from realistic attack scenarios in a smart environment. The data includes both normal and

¹<https://www.kaggle.com/datasets/alaaelmor/ton-iot-train-test-network>

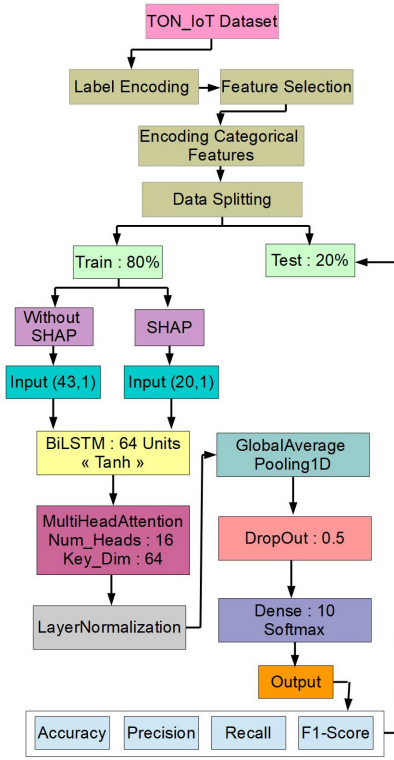


Fig. 1: BiLSTM-MHA Architecture for Explainable IoT IDS.

malicious network activities, encompassing a wide variety of cyberattacks, including: DoS, DDoS, scanning, injection attacks, backdoors, password attacks, ansomware, XSS, MITM and normal traffic [14].

B. Evaluation Metrics for the TON_IoT Dataset

In order to assess the suggested intrusion detection model's efficacy, several standard classification metrics are employed, including Accuracy, F1-Score, Precision, and Recall. These metrics are widely used in the literature to assess the effectiveness of machine learning (ML) and DL models in cyber threat detection and classification tasks. Their definitions and applications have been thoroughly discussed in numerous prior studies [15–19]. These metrics offer a thorough understanding of the model's capacity to accurately distinguish between malicious and legitimate traffic, especially when multi-class categorization is involved.

C. Experiments and Results

The experimental evaluation was conducted using the Google Colab environment, leveraging its GPU capabilities for efficient model training. This section highlights the usefulness of the suggested architecture in multi-class intrusion detection tasks by presenting its performance results on both standard test data and zero-day attack scenarios.

- **Accuracy and loss graph:** Figure 2 (a) and (b) illustrates the accuracy and loss curves, for both training and validation sets across 20 epochs. From the accuracy plot

in Figure 2 (a), it is evident that the model achieves rapid convergence within the first few epochs. Specifically, there is a steep rise in both training and validation accuracy between epochs 1 and 3, indicating that the model quickly learns meaningful representations from the data. After epoch 3, the accuracy stabilizes and fluctuates only slightly, consistently maintaining a high value above 98.79% for validation sets. This convergence suggests that the BiLSTM-MHA model is able to generalize well, with no signs of overfitting or underfitting throughout the training process. The loss curves in Figure 2 (b) show a complementary trend to the accuracy plots. There is a significant drop in both training and validation loss during the first few epochs, particularly between epochs 1 and 3. This indicates effective learning and minimization of error. After epoch 3, the loss values plateau, approaching values near 0.0391, which supports the observation of strong convergence and stable performance.

- **Discussion:** The close alignment between training and validation curves in both loss and accuracy demonstrates that the model maintains a good balance between bias and variance. The absence of divergence between these curves implies that the training data is not being overfitted by the model, and its performance is robust on unseen validation samples. This highlights the effectiveness of integrating BiLSTM layers with MHA, which likely enhances the model's capability to capture temporal dependencies and contextual patterns in the network traffic data.

- **Classification Report:** Table I presents the multiclass classification report of the proposed BiLSTM-MHA model. Overall, the model's accuracy was 99%, with a macro-averaged F1-score of 94% and a weighted F1-score of 99%, indicating strong generalization across both frequent and infrequent classes. Most attack types, including DoS, injection, DDoS, and ransomware, were detected with high precision and recall, typically above 94%. Notably, the model achieved perfect classification for the Backdoor class, with 100% precision, recall, and F1-score. Similarly, the Normal traffic and Password classes were detected with near-perfect scores, underscoring the model's robustness in distinguishing benign from malicious traffic. However, performance dropped significantly for the MITM (Man-in-the-Middle) class, with a precision of 70%, recall of 54%, and an F1-score of 61%. This decline can be attributed to the severe class imbalance, as the MITM class contains only 267 instances out of 89,995. These findings demonstrate the BiLSTM-MHA architecture's general efficacy in multiclass intrusion detection, while also suggesting that handling rare classes remains a challenge requiring further investigation.

- **Confusion matrix:** The normalized confusion matrix of the BiLSTM-MHA model assessed on the validation set is shown in Figure 3. The model's classification performance across the ten attack types from the TON_IoT dataset is clearly represented by the matrix. The confusion matrix's

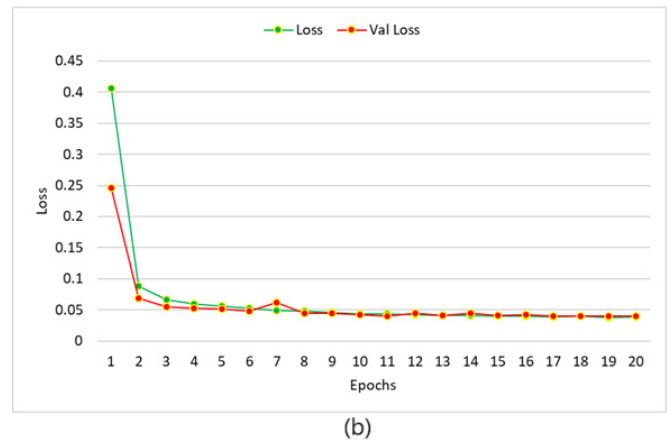
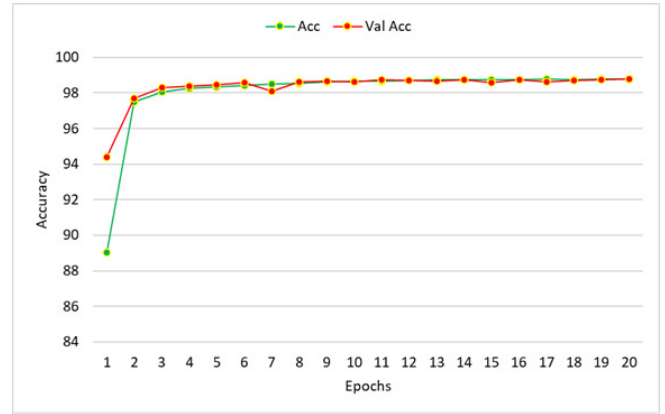


Fig. 2: Accuracy and loss trends of the Proposed BiLSTM-MHA Architecture: (a) Accuracy, (b) Loss.

TABLE I: Multiclass classification report for the proposed BiLSTM-MHA model.

	Precision	Recall	F1-score	Support
Benign traffic (0)	99%	100%	100%	57674
Scanning (1)	99%	98%	99%	3960
DoS (2)	96%	100%	98%	3887
Injection (3)	94%	98%	96%	3829
DDoS (4)	97%	97%	97%	3925
Password (5)	98%	100%	99%	3908
XSS (6)	99%	94%	96%	4189
Ransomware (7)	99%	94%	96%	4310
Backdoor (8)	100%	100%	100%	4046
MITM (9)	70%	54%	61%	267
Accuracy			99%	89995
Macro avg	95%	93%	94%	89995
Weighted avg	99%	99%	99%	89995

diagonal entries show the percentage of samples for each class that were properly classified. The model achieves remarkably high accuracy for most classes: Normal, Scanning, XSS, Ransomware, and Backdoor are classified with nearly perfect accuracy (equal or more than 99%). Classes such as DoS, DDoS, Password, and injection also show strong performance with accuracies ranging between 94% and 98%. This high diagonal dominance reflects

the robustness of the model in differentiating between various network activities and attack patterns. A notable exception is the MITM class, where the model achieves only 70% accuracy. Furthermore, 24% of MITM instances are misclassified as Normal traffic. This reduced performance is primarily attributed to the severe class imbalance in the dataset: the MITM class contains a significantly lower number of samples compared to other attack types which contains the same number of samples, while the Normal class dominates the dataset in volume. Due to the limited representation of MITM samples during training, the model is likely underexposed to its distinguishing patterns, causing it to generalize poorly for this minority class. Furthermore, the semantic resemblance between innocuous traffic and MITM behavior may make it more difficult for the model to discern between the two.

- Discussion: The confusion matrix confirms the overall robustness and high precision of the proposed model across most classes. However, the misclassification of MITM traffic underscores the need for enhanced feature engineering, additional data augmentation, or class-specific fine-tuning strategies to improve detection of more subtle or stealthy threats.

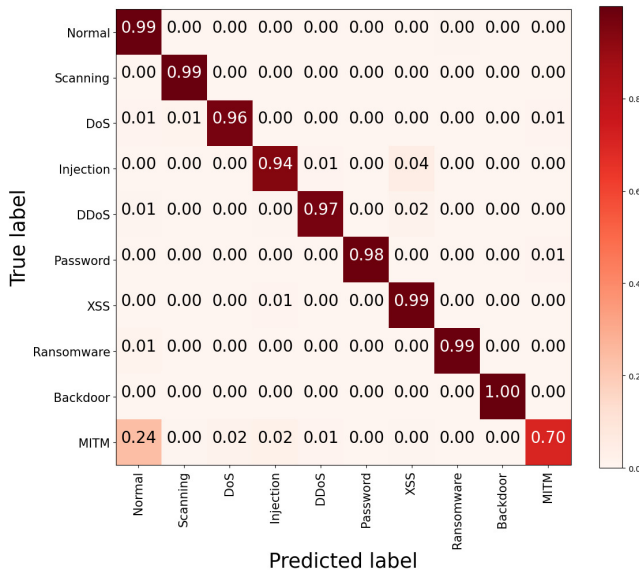


Fig. 3: Confusion matrix of the proposed BiLSTM-MHA model.

- **Receiver operating characteristics (ROC):** Figure 4 presents the ROC curves for the ten-class classification problem, where each curve corresponds to one class in the TON_IoT dataset. The Area Under the Curve (AUC) values are also reported for each class, providing a quantitative measure of the model's discriminative ability. All ten ROC curves (representing classes 0 through 9) exhibit near-perfect performance with AUC values equal to 1.00, indicating that the BiLSTM-MHA model achieves excellent separability across all classes. This means the model is highly capable of distinguishing between positive

and negative instances for each individual class, regardless of threshold.

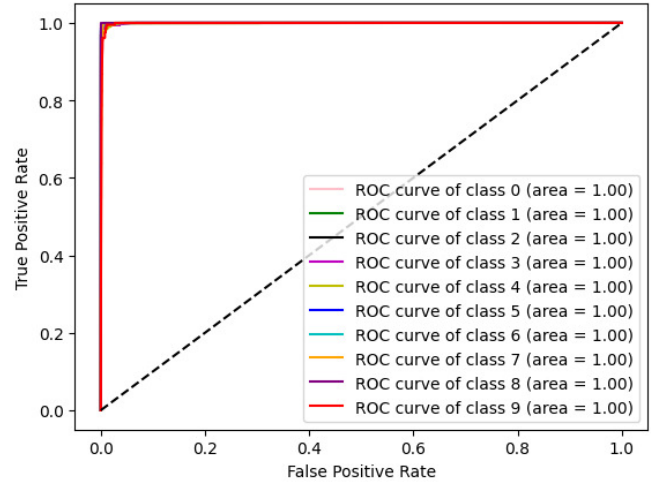


Fig. 4: ROC curve of the proposed BiLSTM-MHA model.

- **Comparison with state-of-the-art:** Table II evaluates the performance of the suggested BiLSTM-MHA model using the TON_IoT dataset and contrasts it with a number of current methods, such as Random Forest (RF), Convolutional Neural Network (CNN), XGBoost, and Transformer-based models.

The suggested BiLSTM-MHA model performs better than the others on every important evaluation metric. With an impressively low loss of 0.0391 and an inference time of 0.0002 seconds per instance, it attains the highest accuracy of the compared methods (98.79%), demonstrating robust generalization and efficient convergence. Furthermore, it attains the lowest **FPR** of 0.0046%, a crucial factor in intrusion detection systems (IDS) to minimize false alarms and ensure reliable threat detection. Regarding prior studies, XGBoost [20], although relatively strong with 98.2% accuracy and similar precision/recall, it still underperforms compared to BiLSTM-MHA in terms of accuracy and FPR (0.009 vs. 0.0046). DL alone (CNN [22] /Transformer [23]) does not guarantee high performance; the combination of BiLSTM with MHA appears to be key in capturing both sequential and contextual information in the TON_IoT data. In addition, many previous studies do not report all metrics (e.g., loss, FPR), which limits their reproducibility and transparency. In contrast, the proposed method provides a complete set of performance metrics. Overall, the comparative results in Table II clearly demonstrate that the BiLSTM-MHA model outperforms existing state-of-the-art models for multiclass intrusion detection on the TON_IoT dataset. Its high accuracy, low loss, excellent precision/recall, and notably low FPR highlight its potential as a robust and reliable model for real-world cybersecurity applications.

TABLE II: Evaluation of Classification Metrics for the Proposed BiLSTM-MHA Model Compared to Leading Models in Multiclass Contexts.

Work	year	Model	datasets	Acc. (%)	Loss	Pr. (%)	Rc (%)	F1 (%)	FPR (%)	Inf time (Sec/inst)
A. R. Gad et al. [20]	2021	XGboost	TON_IoT	98.2	$\times$	98.2	98.2	98.2	0.009	$\times$
E. M. Maseno et al. [21]	2024	RF	TON_IoT	98	$\times$	$\times$	$\times$	$\times$	$\times$	$\times$
B. N. Shaker et al. [22]	2023	CNN	TON_IoT	69.08	$\times$	$\times$	$\times$	61	$\times$	$\times$
S. S. Almalki et al. [23]	2025	Transformer	TON_IoT	68.3	$\times$	$\times$	$\times$	61	1.9	$\times$
Presented	2025	BiLSTM-MHA	TON_IoT	98.79	0.0391	98.88	98.76	98.78	0.0046	0.0002
			EdgeIoT	99.02	0.0261	99.03	99.02	99.02	–	0.0001

D. Zero-Day Attack Detection Assessment

Using a leave-one-attack-out zero-Day strategy (LOAO), a zero-day attack experiment was carried out to assess the suggested model's generalizability to yet undiscovered attack types. Specifically, the TON_IoT dataset was used, and one attack class "Class 2 which represent the scanning attack" was deliberately excluded from the training phase. After being trained on the remaining attack and benign classes, the model was evaluated on the Class 2 samples that were excluded, creating a realistic zeroday scenario. The model's validation accuracy during training was 98.73%, and its validation loss over 10 epochs was 0.0415, indicating strong learning performance and convergence. In the zero-day testing phase, where Class 2 was introduced for the first time, the model demonstrated robust generalization capabilities by achieving a zero-day testing accuracy of 97.90% and a test loss of 0.0591. Despite never encountering this attack type during training, the model was able to accurately identify the anomalous behavior associated with the new class. Furthermore, the performance on key evaluation metrics during zero-day testing was highly promising: Pr: 100%, Rc: 97.90%, and a F1-Score: 98.94%. These findings imply that the model may recognize and extrapolate unknown attack patterns in addition to learning discriminative features from known attack classes. The high precision indicates that all detections of the zero-day attack were indeed correct (i.e., no false positives), while the near-perfect recall and F1-score underscore the model's strong balance between sensitivity and accuracy. Overall, the experiment demonstrates that the proposed model generalizes well to new and unknown attack types, a critical capability for IDS in real-world and evolving threat environments. This ability to detect zero-day attacks highlights the effectiveness of the model architecture and the feature representation learned from the TON_IoT dataset.

E. BiLSTM-MHA model explainability

To improve the proposed BiLSTM-MHA model's transparency, the contribution of each unique input feature to the model's predictions was interpreted using SHAP (SHapley Additive exPlanations). As shown in Figure 5, the color-coded SHAP bars correspond to different classes, enabling a multi-class interpretability perspective. This visualization not only reveals which features are important globally but also how their impact varies across different attack types.

Feature 0 "ts" and feature 1 "src_ip" exhibit the highest mean SHAP values, indicating they are the most influential

features across multiple classes. Their strong contribution suggests that these variables capture essential characteristics that help differentiate between normal and attack traffic.

Several other features; such as feature 10 "conn_state", Feature 13 "src_ip_bytes", Feature 4 "dst_port", and Feature 12 "src_pkts"; also show moderate influence, but their contributions appear to be more class-specific, affecting certain attack types (e.g., Classes 4, 5, 7, and 8) more than others.

Lower-ranked features (e.g., feature 2 "src_port", 3 "dst_ip", and 7 "duration") contribute minimally to the model's predictions affecting some classes such as class 0, and 2.

Certain features were predominantly associated with specific classes; for example, features 5 "proto", 6 "service", and 22 "dns_RA" contributed primarily to classes 0, 5, and 2, respectively

- Discussion: By integrating SHAP into the evaluation, this study moves beyond traditional black-box neural network outputs and offers transparent, explainable AI (XAI). The SHAP analysis confirms that the model's decisions are not arbitrary; rather, they are grounded in consistent, identifiable feature patterns. This is particularly valuable in IDS, where explainability is essential for security analysts to validate alerts, understand attack vectors, and ensure trust in AI-assisted decision-making.

F. SHAP-Based Dimensionality Reduction and Its Impact on Model Performance

To further improve the suggested IDS framework's efficiency and interpretability, SHAP was employed (Figure 5) not only for post hoc explanation but also as a feature selection mechanism. By analyzing the global feature importance values derived from SHAP, the top 20 most influential features (Table III) were identified and retained for training the final version of the model, effectively reducing the input dimensionality from 43 to 20. This explainability-driven reduction significantly decreased the model's complexity while preserving performance metrics. After retraining the BiLSTM-MHA model using only the selected 20 features, we have achieved an accuracy and loss of 98.78% and 0.0395 respectively, with an F1-score of 98.75% and Pr and Rc of 98.79% and 98.73%, respectively. The inference time achieved is 0.0002 seconds per instance. These results highlight that a reduced feature set, informed by SHAP explanations, can maintain near-identical predictive power while improving computational efficiency. This not only

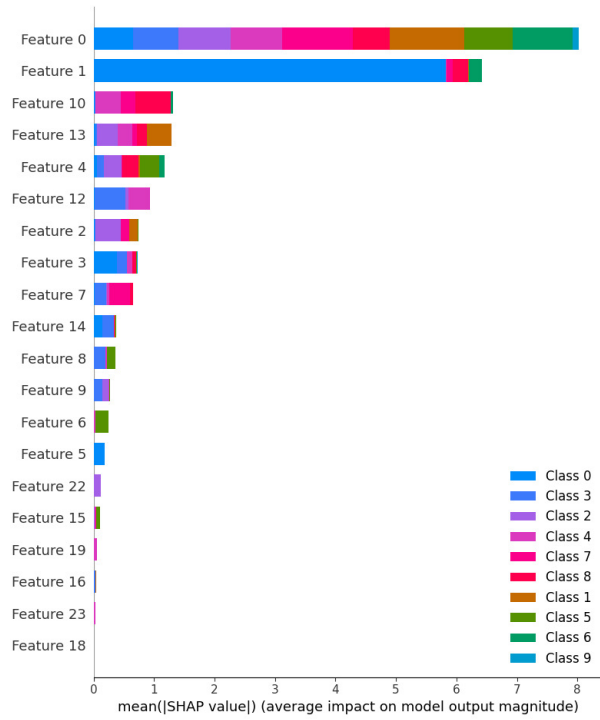


Fig. 5: Average SHAP Values Indicating Feature Impact on Model Output Magnitude.

supports faster inference, which is crucial in real-time IoT environments, but also confirms that many of the original 43 features contribute minimally to the detection task and can be safely excluded. Ultimately, this demonstrates the dual benefit of SHAP in the context of IDS: enhancing both explainability and operational performance.

TABLE III: Features Employed in SHAP Analysis with Their Corresponding Names.

Feature number	Feature name	Feature number	Feature name
0	ts	8	src_bytes
1	src_ip	9	dst_pkts
10	conn_state	6	service
13	src_ip_bytes	5	proto
4	dst_port	22	dns_RA
12	src_pkts	15	dst_ip_bytes
2	src_port	19	dns_rcode
3	dst_ip	16	dns_query
7	duration	23	dns_rejected
14	dst_pkts	18	dns_qtype

G. Ablation Study

Table IV presents the ablation study results for different configurations of the proposed BiLSTM-MHA model in the multiclass classification task. The goal of this analysis is to assess the influence of the number of attention heads, dropout rate, and the application of the focal loss (FL) technique on the model's performance in terms of accuracy, loss, FPR, and inference time.

As shown in Cases #1–#4, increasing the number of attention heads generally improves performance. When the number of attention heads is increased from 2 to 16, the accuracy improves from 98.00% to 98.79%, while the loss decreases from 0.0677% to 0.0391%. The FPR also slightly decreases, indicating a reduction in misclassifications.

Comparing Cases #4, #5, and #6 (all with 16 attention heads), we can observe the effect of varying the dropout rate. A moderate dropout rate of 0.5 (Case #4) yields the best generalization performance, achieving 98.79% accuracy and the lowest loss of 0.0391%.

In Case #7, the BiLSTM-MHA model with Focal Loss and 16 attention heads achieves the best overall results, with an accuracy of 98.81%, the lowest loss of 0.0040%, and an FPR of 0.0036%. These improvements demonstrate that Focal Loss effectively addresses class imbalance and enhances model robustness by focusing more on hard-to-classify examples. The inference time remains comparable to the standard configuration (0.00018 sec per inst), showing that the performance gain does not significantly increase computational cost.

TABLE IV: Performance of different variants of the proposed models in multiclass classification.

Case	Model	N.of heads	Dropout	Acc	Loss	FPR	Inf. time
#1	BiLSTM-MHA	2	0.5	98.00	0.0677	0.0069	0.00011
#2	BiLSTM-MHA	4	0.5	98.55	0.0473	0.0037	0.00011
#3	BiLSTM-MHA	8	0.5	98.63	0.0432	0.0032	0.00017
#4	BiLSTM-MHA	16	0.5	98.79	0.0391	0.0046	0.0002
#5	BiLSTM-MHA	16	0.3	98.63	0.0429	0.0038	0.00014
#6	BiLSTM-MHA	16	0.2	98.51	0.0484	0.0061	0.00014
#7	BiLSTM-MHA with FL tech	16	0.5	98.81	0.0040	0.0036	0.00018

V. CONCLUSION

In this work, we presented an effective and interpretable IDS tailored for the unique challenges of IoT environments. By leveraging a hybrid BiLSTM-Multi-Head Attention (BiLSTM-MHA) architecture, our model successfully captured complex temporal dependencies and enhanced feature interactions, leading to highly accurate intrusion detection. Experimental evaluations on the TON_IoT dataset demonstrated strong performance, achieving an accuracy of 98.79%, low false positive rate (0.0391%), and minimal inference latency (0.002 seconds per instance), which is suitable for real-time IoT applications. Beyond performance, we addressed the interpretability of the model by integrating SHAP-based Explainable AI techniques. This enabled a transparent understanding of the decision-making process and highlighted the contribution of important features across various attack classes, providing valuable insights for cybersecurity practitioners. To further validate the generalization capabilities of the model, we conducted a Zero-Day attack assessment using a Leave-One-Attack-Out (LOAO) strategy. The model demonstrated strong robustness by accurately detecting previously unseen attacks, achieving a zero-day testing accuracy of 97.91% and an F1-score of 98.94%.

The proposed BiLSTM-MHA-based IDS offers a comprehensive solution that combines high detection performance, efficiency, generalizability, and interpretability; making it a strong candidate for deployment in real-world IoT security systems. Future work may explore optimizing the model for edge devices, extending it to federated learning frameworks, and incorporating adaptive mechanisms to respond to evolving threat landscapes.

REFERENCES

- [1] Y. Otoum, D. Liu, and A. Nayak, "Dl-ids: a deep learning-based intrusion detection framework for securing iot," *Transactions on Emerging Telecommunications Technologies*, vol. 33, no. 3, p. e3803, 2022.
- [2] A. Mall, P. Singh, A. Thute, S. P. Khapre, and A. Shankar, "Security issues of edge computing in iot," in *Proceedings of International Conference on Machine Intelligence and Data Science Applications: MIDAS 2020*. Springer, 2021, pp. 567–579.
- [3] I. Idrissi, M. Azizi, and O. Moussaoui, "A lightweight optimized deep learning-based host-intrusion detection system deployed on the edge for iot," *International Journal of Computing and Digital System*, 2022.
- [4] K. Bartos, M. Sofka, and V. Franc, "Optimized invariant representation of network traffic for detecting unseen malware variants," in *25th USENIX Security Symposium (USENIX Security 16)*, 2016, pp. 807–822.
- [5] I. Idrissi, M. Azizi, and O. Moussaoui, "Accelerating the update of a dl-based ids for iot using deep transfer learning," *Indones. J. Electr. Eng. Comput. Sci*, vol. 23, no. 2, pp. 1059–1067, 2021.
- [6] L. Sana, M. M. Nazir, J. Yang, L. Hussain, Y.-L. Chen, C. S. Ku, M. Alatiyyah, and L. Y. Por, "Securing the iot cyber environment: Enhancing intrusion anomaly detection with vision transformers," *IEEE Access*, 2024.
- [7] S. Racherla, P. Sripathi, N. Faruqui, M. A. Kabir, M. Whaiduzzaman, and S. A. Shah, "Deep-ids: A real-time intrusion detector for iot nodes using deep learning," *IEEE Access*, vol. 12, pp. 63 584–63 597, 2024.
- [8] B. Nawaal, U. Haider, I. U. Khan, and M. Fayaz, "Signature-based intrusion detection system for iot," in *Cyber security for next-generation computing technologies*. CRC Press, 2024, pp. 141–158.
- [9] Z. Abou El Houda, B. Brik, and L. Khoukhi, "“why should i trust your ids?”: An explainable deep learning framework for intrusion detection systems in internet of things networks," *IEEE Open Journal of the Communications Society*, vol. 3, pp. 1164–1176, 2022.
- [10] A. Gueriani, H. Kheddar, A. C. Mazari, and M. C. Ghanem, "A robust cross-domain ids using bigru-lstm-attention for medical and industrial iot security," *ICT Express*, 2025.
- [11] A. Alqahtani, "Optimized deep autoencoder and bilstm for intrusion detection in iots-fog computing," *Multimedia Tools and Applications*, vol. 84, no. 8, pp. 4907–4943, 2025.
- [12] S. Bi, J. Wang, J. Song, P. Li, and L. Li, "Research on the intrusion detection model for power internet of things combining deep belief network and bilstm," *Journal of Cyber Security and Mobility*, vol. 14, no. 3, pp. 653–672, 2025.
- [13] R. Nagarajan *et al.*, "A novel bidirectional lstm model for network intrusion detection in sdn-iot network," *Computing*, vol. 106, no. 8, 2024.
- [14] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "Ton_iot telemetry dataset: A new generation dataset of iot and iiot for data-driven intrusion detection systems," *Ieee Access*, vol. 8, pp. 165 130–165 150, 2020.
- [15] A. Gueriani, H. Kheddar, and A. C. Mazari, "Deep reinforcement learning for intrusion detection in IoT: A survey," in *2023 2nd International Conference on Electronics, Energy and Measurement (IC2EM)*, vol. 1. IEEE, 2023, pp. 1–7.
- [16] H. Kheddar, M. Hemis, Y. Himeur, D. Megías, and A. Amira, "Deep learning for steganalysis of diverse data types: A review of methods, taxonomy, challenges and future directions," *Neurocomputing*, p. 127528, 2024.
- [17] H. Kheddar, Y. Himeur, S. Al-Maadeed, A. Amira, and F. Bensaali, "Deep transfer learning for automatic speech recognition: Towards better generalization," *Knowledge-Based Systems*, vol. 277, p. 110851, 2023.
- [18] A. Gueriani, H. Kheddar, and A. C. Mazari, "Enhancing iot security with cnn and lstm-based intrusion detection systems," in *2024 6th International Conference on Pattern Analysis and Intelligent Systems (PAIS)*. IEEE, 2024, pp. 1–7.
- [19] —, "Adaptive cyber-attack detection in iiot using attention-based lstm-cnn models," in *2024 International Conference on Telecommunications and Intelligent Systems (ICTIS)*. IEEE, 2024, pp. 1–6.
- [20] A. R. Gad, A. A. Nashat, and T. M. Barkat, "Intrusion detection system using machine learning for vehicular ad hoc networks based on ton-iot dataset," *IEEE access*, vol. 9, pp. 142 206–142 217, 2021.
- [21] E. M. Maseno, Z. Wang, and Y. Sun, "Performance evaluation of intrusion detection systems on the ton_iot datasets using a feature selection method," in *Proceedings of the 2024 8th International Conference on Computer Science and Artificial Intelligence*, 2024, pp. 607–613.
- [22] B. N. Shaker, B. Q. Al-Musawi, and M. F. Hassan, "A comparative study of ids-based deep learning models for iot network," in *Proceedings of the 2023 International Conference on Advances in Artificial Intelligence and Applications*, 2023, pp. 15–21.
- [23] S. S. Almalki, "A deep learning-based framework for real-time detection of cybersecurity threats in iot environments," *International Journal of Advanced Computer Science & Applications*, vol. 16, no. 3, 2025.